

1. Introduction

Retail demand forecasting plays a critical role in inventory optimization and supply chain efficiency.

This project builds a multi-store sales forecasting system using machine learning techniques.

The objective is to reduce forecast error and provide measurable business impact.

2. Business Problem Definition

Fashion retail operates in a fast-changing demand environment.

Inaccurate forecasts lead to stockouts or excess inventory.

The goal is to predict daily store-level sales to improve planning accuracy.

3. Dataset Description

The dataset includes daily sales data across multiple stores.

Key variables include date, store number, product family, sales, and promotion count.

Sales were aggregated at the store-day level for forecasting.

4. Exploratory Data Analysis

EDA revealed strong weekly seasonality and trend patterns.

Promotional activity showed clear correlation with sales spikes.

Store-level variability justified multi-store modeling.

5. Feature Engineering

Time-based features: month, day of week, week of year, year.

Lag features: lag_1, lag_7, lag_30 to capture temporal dependency.

Rolling averages: 7-day and 30-day windows to capture trends.

6. Modeling Strategy

A time-based train-test split was used to avoid leakage.

Baseline model: naive lag-1 forecast.

Machine learning models: Random Forest and XGBoost.

7. Model Evaluation

Baseline RMSE: 5348.

Random Forest RMSE: 2485.

XGBoost RMSE: 2278.

MAPE: 9.75%.

XGBoost reduced RMSE by 57% compared to baseline.

8. Business Impact Analysis

Sub-10% MAPE indicates production-level forecasting capability.

Improved forecast accuracy reduces safety stock requirements.

Better demand prediction enhances operational efficiency.

9. Feature Importance Insights

Lag features were the strongest predictors, confirming temporal dependence.

Weekly seasonality played a major role in demand fluctuations.

Promotions contributed significantly to sales variability.

10. Conclusion & Future Work

The forecasting pipeline demonstrates strong predictive performance.

Future work could include LSTM models and hierarchical forecasting.

Integration into inventory optimization systems would enhance impact.